

Leo Wang

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Education

2024-27 ♦ **University of Maryland College Park, Bachelor of Science.**

- Double Major: **Computer Science** (Data Science) and **Math** with a **4.0** GPA.
- Computer Science Departmental Honors Program.
- Dean's list (all semesters).

2024-27 ♦ **UMD Presidential Scholarship** (\$5,000/year)

2024-27 ♦ **National Merit Scholarship** (\$1,000/year)

Experiences

Aug. 25 – ♦ **TestuGo: Bluetooth Navigation and Exploration Application.** Software Engineer, UMD

- Developing a location-based Android and IOS application with Kotlin and Swift to provide information on nearby locations.
- Integrates with ArcGIS to automatically detect when a user enters a building or is nearby dynamic events.
- Helps students navigate campus through hand crafted scavenger hunts and tours.

Jun. – Aug. 25 ♦ **NOAA STAR GNSS-RO Page Development.** Web Developer, CISESS (MD)

- Aided NOAA efforts though a paid internship by extending the functionality of the GNSS-RO page in Python, JavaScript, PHP.
- Analyzed over **1,600,000+** log lines to understand customers, including location, nationality, and trends over time.
- Developed Ground Tracker application to dynamically visualize GNSS-RO profiles in an interactable globe and map projection.
- Replaced previous tooling via newly introduced features, including filtering by mission/GNSS, various color coding options, dynamic time range, built-in screenshotter, permalink system, and more.
- Increased rendered GNSS-RO profiles by **1,600x** (up to **50,000+** profiles in a browser environment).
- Awarded Best CISESS Summer Intern Presentation.

Oct. 24 – Jan. 25 ♦ **Cybersecurity Training Platform for MITRE Corporation.** Web Developer, UMD

- Developed a versatile training platform for MITRE Corporation as part of the App Development Club.
- Wrapped the WebGoat program with a custom UI, user system to track individual tasks, and admin management controls.
- Personally built all the frontend sections of the admin-side pages using NextJS, TailwindCSS and TypeScript.

Projects

Jan. – Aug. 25 ♦ **Radiology Pronunciation Dictionary for Foreign Speakers.** React, TypeScript, Python

- Commissioned to create a personalized and feature-rich phonetic dictionary to help foreign speakers learn English.
- Integrated with the Merriam-Webster dictionary to automatically find word definitions.
- Implemented an overlaying internal dictionary to maximize user control, with **10,000+** words that can be updated in-program.
- Lexicographic engine breaks down all words into their phonetic and stressed components, allowing users to match by sound.

Aug. 24 ♦ **N-Body (Barnes-Hut) Simulation.** C++, Dear ImGui

- Developed an optimized N-body simulation (tested with up to **15,000 bodies**) by leveraging the Barnes-Hut algorithm through recursively dividing the coordinate space into a quadtree structure to achieve an $O(n \log n)$ time complexity (naive is $O(n^2)$).
- Integrated additional optimizations such as leapfrog integration (an efficient numerical method to integrate differential equations), tail recursion, caching, and incremental center of mass calculations.

Skills

Languages ♦ **Advanced:** TypeScript/JavaScript, HTML/CSS, Python
Intermediate: C#, Java, C/C++, PHP, Kotlin

Technologies ♦ **Platforms:** Unity, NextJS
Libraries: TailwindCSS, React.js, REST, Vite, SQL, Bash, Node, Numpy, Selenium, Tensorflow, Matplotlib, SciPy, Pandas, Newtonsoft, Dear ImGui, ImPlot, OpenCV
Dev. Tools: Visual Studio, Android Studio, Visual Studio Code, Git, Github

Skills ♦ Data Visualization, Web Development, Software Development, Linear Algebra